

Chapter 2: Literature Review

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LEO Low Earth Orbit
TLE Two Line Elements
JSPOC Joint Space Operations Center
RMS Root Mean Square
EKF Extended Kalman Filter
UKF Unscented Kalman Filter
UPF Unscented Particle Filter
SRIF Square Root Information Filter
GNSS Global Navigation Satellite System
MAGSAT Magnetic Field Satellite
LACE Low-Power Atmospheric Compensation Experiment
DE-2 Dynamics Explorer-2
BLS Batch Least Squares Filter
ILS Iterated Least Squares Filter
ULS Unscented Least Squares Filter
HLS Hybrid Least Squares Filter
GAN Generative Adversarial Network
ML Machine Learning
CNN Convolutional Neural Network

I. INTRODUCTION

Space has been at the forefront of research and exploration for decades and much of modern technology has come to rely on the innovations and architecture that have been built to maintain it. The largest growing component of space innovation involves the deployment of Low Earth Orbit (LEO) satellites [1]. This greatly expanding front of new technology and ideas brings along its own challenges that range from security to new ways to track orbits [2, 3].

II. CUBESATS

The use of CubeSats has continued to grow in the LEO environment(1) and has become a major component of the space industry [1]. CubeSats are small satellites that are typically built in a standard size and form factor, which allows for easy deployment and integration into existing satellite constellations. This standardization has led to the rapid growth of CubeSat technology including the development of new communication systems [4, 5] and onboard processing capabilities [6]. These developments have enabled CubeSats to perform a wide range of missions, from Earth observation, scientific research, and commercial applications.

A. Security

With the rapid growth of CubeSat technology, there has been an increased focus on the vulnerabilities and security

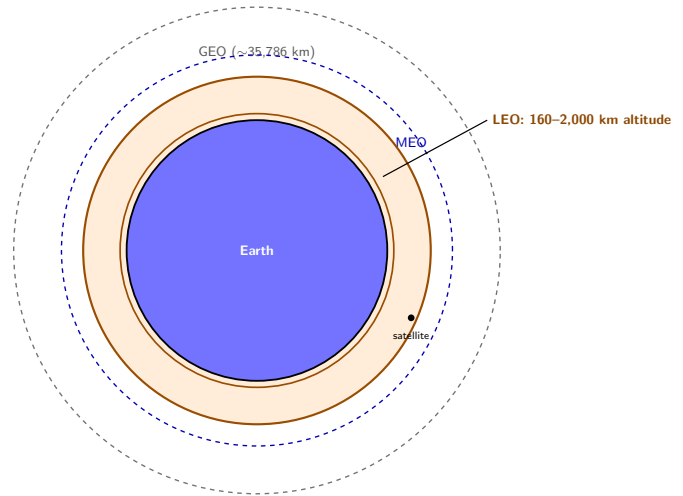


Fig. 1. Low Earth orbit (LEO) altitude range relative to MEO and GEO.

risks associated with these small satellites. Much of these now deployed CubeSats are built with a focus on cost and size much like IoT devices, which has led to a lack of security features and protocols [7]. This makes CubeSats a prime target for research into security and resiliency of these systems.

Ahmad et al. [8] pursue the one tactic of utilizing a Generative Adversarial Network (GAN), a type of neural network, to generate synthetic data to train a Machine Learning (ML) model to detect and mitigate cyber attacks on CubeSats. It utilized the Edge-IIoTset to train the GAN to generate the synthetic data and then used the generated data to train a Convolutional Neural Network (CNN) plus an autoencoder to detect cyber attacks on IoT devices specifically aimed at CubeSats. The results displayed a validation accuracy of 96.56% for the CNN in the first epoch showing that the GAN was able to generate realistic data that could be used to train a ML model to detect cyber attacks on CubeSats. This demonstrates the potential of using GANs to enhance the security of CubeSats and other small satellites.

In the offensive space security domain Calabrese et al. [9] explores the use of a hosted payload to perform a cyber attack against a CubeSat in a simulated environment. The study provides a full demonstrable attack chain, inspired by the SPARTA Matrix [10], the only uses open source tools and software to demonstrate the vulnerabilities of CubeSats and the potential for cyber attacks against them.

TABLE I
TWO-LINE ELEMENT SET FIELD DEFINITIONS

Line	Field	Content
Title	–	Satellite Name
1	1	Line Number
1	2	Satellite Catalog Number
1	3	Classification (U/C/S)
1	4	International Designator (Launch Year)
1	5	International Designator (Launch Number)
1	6	International Designator (Launch Piece)
1	7	Epoch Year
1	8	Epoch (Day and Fractional Day)
1	9	First Derivative of Mean Motion (rev/day ²)
1	10	Second Derivative of Mean Motion (rev/day ³)
1	11	B* Drag Term (Earth radii ⁻¹)
1	12	Ephemeris Type
1	13	Element Set Number
1	14	Checksum (modulo 10)
2	1	Line Number
2	2	Satellite Catalog Number
2	3	Inclination (deg)
2	4	Right Ascension of Ascending Node (deg)
2	5	Eccentricity
2	6	Argument of Perigee (deg)
2	7	Mean Anomaly (deg)
2	8	Mean Motion (rev/day)
2	9	Revolution Number at Epoch
2	10	Checksum (modulo 10)

III. NAVIGATION

LEO satellite use a variety of methods to assist in their own navigation and an organizations ability to track them from the ground. The most common methods used to track a satellite from their ground is through Two Line Elements (TLE)s. TLEs describe the orbital parameters, as seen in Table III, that are required for the tracking and locating of specific satellites during orbit [11]. They provide the specific time and location that each LEO satellite occupies to facilitate communication and to ensure that no satellite impacts another. The recordings of a satellites orbit are taken through observation at Joint Space Operations Center (JSPOC) and provide consistent snapshot updates of currently recorded TLE's every 8 hours [11]. A majority of these satellites have a mean motion of 90 minutes with an altitude(Circular) of 281.49 km [12]. This provides only a few minutes in which the satellite can be contacted before it is out of range and continues until its next orbit. These minimal contact intervals make it vitally important to accurately predict when contact can be made and what communications are of import to send during this window. This creates a need for accurate truly autonomous navigation and orbit determination that does not rely on outside systems to provide accurate information [13].

IV. ALTERNATIVE NAVIGATION

Alternative methods of navigation and orbit determination continue to be explored to provide more robust and accurate systems. A newly explored source of navigation is the use of satellite sensors, like magnetometer and gravimeter, in

conjunction with accurate modeling to derive orbit parameters without specific ground station interference [14–16]. These alternative methods provide an independent system of determination without reliance on outside systems [16].

TABLE II
POSITION ERROR RESULTS FOR THE SRIF AND BATCH LS FILTERS [13]

S/C	Filter	RMS/Total	Peak	Along/Cross-Track
MAGSAT	SRIF	–	<8 km	slow drift / periodic
MAGSAT	Batch LS	4 km RMS	<8 km	3.4 / 1.8 km RMS
DE-2	SRIF	–	–	not tested
DE-2	Batch LS	–	18 km	8–15 km / steady osc.
LACE	SRIF	–	–	not tested
LACE	Batch LS	88 km RMS	125 km	– / dominant

A. Magnetometer

It has been proven by Psiaki and Martel [17] that 5 of the 6 keplerian elements can be determined using only the magnetic field with the missing element being the longitude of the ascending node. Utilizing the magnetometer, known variables like start position and time of flight along with a series of kalman filters the orbit/position of a satellite can be determined [18, 19].

Pisaki and Martel [13] utilize a Square Root Information Filter (SRIF) and Batch Least Squares Filter (BLS) to determine the position of the satellite using only the magnetometer and adjusted based on the truth data. The SRIF was tested on the Magnetic Field Satellite (MAGSAT) launched in October 1979 equipped with a vector magnetometer, the Dynamics Explorer-2 (DE-2) launched in August 1981 equipped with vector magnetometer, and the Low-Power Atmospheric Compensation Experiment (LACE) launched in February 1990 equipped with a scalar magnetometer. The results from the tests can be seen in Table II where the SRIF was able to provide a position estimate with a peak error of less than 8 km for the MAGSAT and the BLS was able to provide a position estimate with a Root Mean Square (RMS) error of 4 km for the MAGSAT, 18 km for the DE-2, and 88 km for the LACE [13]. This demonstrates that the magnetometer can be used to provide a position estimate with a reasonable accuracy, but it is not as accurate as Global Navigation Satellite System (GNSS) and is dependent on the type of magnetometer used.

A series of kalman filters have been explored by Ning and Ma [18] to determine the method and accuracy of differing strategies, this included Unscented Kalman Filter (UKF), Extended Kalman Filter (EKF), and Unscented Particle Filter (UPF). A best case RMS accuracy of 203.32 meters for an EKF, a RMS accuracy of 161.3 meters for an UKF, and a RMS accuracy of 159.7 meters for an UPF when considering a known orbit and 3 second sample rate to filter to a position [18]. The EKF introduces a larger error due to the neglected non-linearities in the system, while the UPF provides a more accurate result at the cost of increased computational complexity [18]. While this method does not provide the same accuracy of GNSS it does provide an independent method of orbit determination that if given the compute power and time

can provide a redundant system of orbit determination without reliance on outside systems.

TABLE III
FILTER CONVERGENCE AND ORBIT DETERMINATION ACCURACY

Metric	Before drift est.	With drift est.
Radial pos. RMS (m)	10.8	10.4
Along-track pos. RMS (m)	1208.3	677.0
Cross-track pos. RMS (m)	37.9	22.8
Radial vel. RMS (m/s)	1.2	0.80
Along-track vel. RMS (m/s)	0.013	0.012
Cross-track vel. RMS (m/s)	0.044	0.026

TABLE IV
RMS ORBIT DIFFERENCES WITH RESPECT TO EGM2008

Gravity Model	Radial diff. (m)	Along-track diff. (m)	Cross-track diff. (m)
EGM96	1.51	3.47	2.32
JGM3	1.66	56.6	6.92
EIGEN-6C4	0.043	13.05	0.57
GGM05G	0.052	12.15	0.55

B. Gravimeter

A further exploration of alternative navigation methods has been explored by Sun et al [15]. This study explores the use of an Iterated Least Squares Filter (ILS), an Unscented Least Squares Filter (ULS), and the proposed Hybrid Least Squares Filter (HLS) utilizing gravity gradient measurements to determine specific orbital elements of a satellite. These filters were implemented on the Gravity Field and Steady-State Ocean Circulation Explorers 12 hour arc starting from 8 September 2013, 00:00:00 GPS time. This study also makes the assumption that the gravity gradient measurements are optimal and available. A baseline gravity model EGM2008 is used to provide reference and as a comparison to the EGM96, JGM3, EIGEN-6C4, and GGM05G models IV. The result of the study shows the HLS model provides the best results with a RMS error of 10.4 meters in the radial position, 677 meters in the along-track position, and 22.8 meters in the cross-track position (Table III).

C. ML Models

While complex simulations and models (like CHAOS-8 and EGM2008) can provide accurate results, they are computationally expensive and require significant time to run. ML models have been explored as a potential solution to similar problems, as they can provide accurate results with significantly less computational resources and time. They also are able to operate in a more autonomous manner, as they can be trained and then hosted on a satellite without the need for constant communication with ground stations. This allows for more efficient and effective navigation and orbit determination, as the satellite can make decisions and adjustments in real-time without relying on external input.

D. Visual Model

TABLE V
THE INITIAL KEPLER ORBIT ELEMENTS OF THE SATELLITE.

Orbit element	Initial value
Semi-major axis	6988.14 km
Eccentricity	0
Inclination	97.77°
Longitude of the ascending node	318.03°
Argument of perigee	0°
Mean anomaly	359.93°

TABLE VI
COMPARISON OF OAD PERFORMANCE USING IMAGES OF FOUR REGULAR-SHAPED OBJECTS

Average RMS	Ellipse	Ellipsoid	Parallel.	Corner
Position error (m)	7.19×10^1	7.22×10^1	7.23×10^1	7.25×10^1
Velocity error (cm/s)	9.79	9.80	9.80	9.81
Attitude error, yaw (deg)	4.24×10^{-3}	4.25×10^{-3}	4.25×10^{-3}	4.23×10^{-3}
Attitude error, pitch (deg)	3.13×10^{-3}	3.15×10^{-3}	3.16×10^{-3}	3.15×10^{-3}
Attitude error, roll (deg)	4.49×10^{-3}	4.50×10^{-3}	4.51×10^{-3}	4.51×10^{-3}
Gyro drift error (deg/s)	1.31×10^{-2}	1.31×10^{-2}	1.31×10^{-2}	1.31×10^{-2}

Muzi Li and Bo Xu [20] introduced a ML based approach to orbit and attitude determination using regular-shapes of ground objects. The simulations were initialized at 12:00:00 UTC on 1 July 2009, using the orbit elements in Table V with full perturbations and a three-axis stabilized, Earth-oriented attitude mode. The camera has a 60° field of view and 1 m focal length, sampling images and gyro data at 1 Hz. Gaussian zero-mean noise (1σ) was applied to image resolution (10 m), ground object position (10 m), gyro measurement ($\sqrt{10} \times 10^{-10}$ rad/s), and focal length (0.01%, ≈ 0.1 mm). Initial biases of 1 km position, 1 m/s velocity, 0.1° attitude, and 0.2°/h gyro drift were included, with a 50% probability of cloud cover obscuring the imaging region. The results of this experiment (Table VI) show that, the ellipse-shaped ground objects yielded the best overall OAD performance, with the lowest position error (7.19×10^1 m), velocity error (9.79 cm/s), and yaw attitude error (4.24×10^{-3} °) among the four shapes. However, the differences across shapes are marginal, indicating that OAD accuracy is largely insensitive to the specific geometry of the ground object.

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